

Preliminary Identities

Proposition 1.1.

$$I = \int_{-\infty}^{\infty} u^{2k} e^{-\alpha u^2/2} du = \sqrt{2\pi} \alpha^{-(2k+1)/2} \prod_{j=1}^k (2j-1) \quad (1.1)$$

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Proof. Let $K(\alpha, u) = \exp(-\alpha u^2/2)$. Consider first the integral:

$$I_0 = \int_{-\infty}^{\infty} K(\alpha, u) du = \int_{-\infty}^{\infty} e^{-\alpha u^2/2} du.$$

Make the change of variables $\omega = \alpha u^2/2$, with differential:

$$d\omega = \alpha u du \implies du = 2^{-1/2} \alpha^{-1/2} \omega^{-1/2} d\omega.$$

Then, I_0 evaluates to:

$$I_0 = 2 \int_0^{\infty} e^{-\omega} \cdot 2^{-1/2} \alpha^{-1/2} \omega^{-1/2} d\omega = 2^{1/2} \alpha^{-1/2} \int_0^{\infty} \omega^{1/2-1} e^{-\omega} d\omega = 2^{1/2} \alpha^{-1/2} \Gamma(1/2).$$

Overall, the integral of $K(\alpha, u)$ evaluates to:

$$I_0 = \sqrt{2\pi} \alpha^{-1/2}. \quad (1.2)$$

Next consider the partial of K wrt α :

$$\frac{\partial}{\partial \alpha} K(\alpha, u) = \frac{\partial}{\partial \alpha} e^{-\alpha u^2/2} = -\frac{u^2}{2} e^{-\alpha u^2/2} = -\frac{u^2}{2} K(\alpha, u).$$

The k th partial is:

$$\frac{\partial^k}{\partial \alpha^k} K(\alpha, u) = \frac{\partial^k}{\partial \alpha^k} e^{-\alpha u^2/2} = (-1)^k \frac{u^{2k}}{2^k} e^{-\alpha u^2/2} = (-1)^k \frac{u^{2k}}{2^k} K(\alpha, u).$$

Rearranging gives:

$$u^{2k} K(\alpha, u) = (-1)^k 2^k \frac{\partial^k}{\partial \alpha^k} K(\alpha, u).$$

Now integral is expressible as:

$$I = \int_{-\infty}^{\infty} u^{2k} K(\alpha, u) du = 2^k (-1)^k \frac{\partial^k}{\partial \alpha^k} \int_{-\infty}^{\infty} K(\alpha, u) du.$$

Applying (1.1.2):

$$I = 2^k (-1)^k \frac{\partial^k}{\partial \alpha^k} \sqrt{2\pi} \alpha^{-1/2} = 2^{k+1/2} (-1)^k \sqrt{\pi} \frac{\partial^k}{\partial \alpha^k} \alpha^{-1/2}.$$

Taking the derivatives in α :

$$\frac{\partial^k}{\partial \alpha^k} \alpha^{-1/2} = (-1)^k \frac{1}{2} \cdot \frac{3}{2} \cdots \frac{(2k-1)}{2} \alpha^{-(2k+1)/2} = \frac{(-1)^k}{2^k} \alpha^{-(2k+1)/2} \prod_{j=1}^k (2j-1)$$

Substituting the derivative into the expression for I :

$$I = 2^{k+1/2-k} (-1)^{2k} \sqrt{\pi} \alpha^{-(2k+1)/2} \prod_{j=1}^k (2j-1).$$

Simplifying gives the result. ■

Corollary 1.1.

$$\int_{-\infty}^{\infty} u^{2k+1} e^{-\alpha u^2/2} du = 0.$$

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Proof. The integral is expressible as:

$$I = 2^k (-1)^k \frac{\partial^k}{\partial \alpha^k} \int_{-\infty}^{\infty} u e^{-\alpha u^2/2} du = 0.$$

Here the integral evaluates to zero due to odd parity of the integrand about the origin. ■

Laplace Approximation

Result 2.1. Suppose $g(x)$ achieves a global minimum at $x^* \in (a, b)$, then for $n \rightarrow \infty$:

$$\int_a^b e^{-ng(x)} dx = e^{-ng(x^*)} \sqrt{\frac{2\pi}{n\ddot{g}(x^*)}} \left\{ 1 - \frac{1}{8n} \kappa_4 + \frac{5}{24n} \kappa_3^2 + \mathcal{O}(n^{-2}) \right\}. \quad (2.1)$$

where:

$$\kappa_3 = \frac{g^{(3)}(x^*)}{\ddot{g}^{3/2}(x^*)}, \quad \kappa_4 = \frac{g^{(4)}(x^*)}{\ddot{g}^2(x^*)}.$$

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Proof. Let $I = \int_a^b \exp\{-ng(x)\} dx$ denote the integral of interest. Take the Taylor expansion of $g(x)$ about x^* :

$$g(x) = g(x^*) + \frac{1}{2} \ddot{g}(x^*) (x - x^*)^2 + \sum_{k=3}^{\infty} \frac{g^{(k)}(x^*)}{k!} (x - x^*)^k.$$

Substituting the Taylor expansion into I :

$$I = e^{-ng(x^*)} \int_a^b e^{-n\ddot{g}(x^*)(x-x^*)^2/2} \exp \left\{ -n \sum_{k=3}^{\infty} \frac{g^{(k)}(x^*)}{k!} (x-x^*)^k \right\} dx.$$

Let $S_3(x)$ denote the series:

$$S_3(x) = -n \sum_{k=3}^{\infty} \frac{g^{(k)}(x^*)}{k!} (x-x^*)^k.$$

Take the Taylor expansion of the second exponential term:

$$\exp \{S_3(x)\} = \left\{ 1 + S_3(x) + \frac{1}{2} S_3(x)^2 + \cdots \right\},$$

then:

$$I = e^{-ng(x^*)} \int_a^b e^{-n\ddot{g}(x^*)(x-x^*)^2/2} \left\{ 1 + S_3(x) + \frac{1}{2} S_3(x)^2 + \cdots \right\} dx.$$

Make the change of variables $\omega = \sqrt{n\ddot{g}(x^*)}(x-x^*)$ with $d\omega = \sqrt{n\ddot{g}(x^*)}dx$:

$$I = \frac{e^{-ng(x^*)}}{\sqrt{n\ddot{g}(x^*)}} \int_{\sqrt{n\ddot{g}(x^*)(a-x^*)}}^{\sqrt{n\ddot{g}(x^*)(b-x^*)}} e^{-\omega^2/2} \left\{ 1 + S_3(\omega) + \frac{1}{2} S_3(\omega)^2 + \cdots \right\} d\omega,$$

where:

$$S_3(\omega) = -n \sum_{k=3}^{\infty} \frac{g^{(k)}(x^*)}{k!} \frac{\omega^k}{n^{k/2} \ddot{g}(x^*)^{k/2}} = - \sum_{k=3}^{\infty} \frac{g^{(k)}(x^*)}{k!} \frac{\omega^k}{n^{k/2-1} \ddot{g}(x^*)^{k/2}}$$

Expanding $\mathcal{O}(n^{-1})$ terms from $S_3(\omega)$:

$$I = \frac{e^{-ng(x^*)}}{\sqrt{n\ddot{g}(x^*)}} \int_{\sqrt{n\ddot{g}(x^*)(a-x^*)}}^{\sqrt{n\ddot{g}(x^*)(b-x^*)}} e^{-\omega^2/2} \left\{ 1 - \frac{g^{(4)}(x^*)\omega^4}{4!n\ddot{g}(x^*)^2} + \frac{1}{2} \left(\frac{g^{(3)}(x^*)\omega^3}{3!n^{1/2}\ddot{g}(x^*)^{3/2}} \right)^2 + \mathcal{O}(n^{-2}) \right\} d\omega,$$

where odd powers of ω are neglected, since these will integrate to zero. For $n \rightarrow \infty$, the bounds of integration may be approximated as infinite:

$$I = \frac{e^{-ng(x^*)}}{\sqrt{n\ddot{g}(x^*)}} \int_{-\infty}^{\infty} e^{-\omega^2/2} \left\{ 1 - \frac{g^{(4)}(x^*)\omega^4}{4!n\ddot{g}(x^*)^2} + \frac{1}{2} \left(\frac{g^{(3)}(x^*)\omega^3}{3!n^{1/2}\ddot{g}(x^*)^{3/2}} \right)^2 + \mathcal{O}(n^{-2}) \right\} d\omega.$$

Applying (1.1.1) to integrate the even powers of ω :

$$I = e^{-ng(x^*)} \sqrt{\frac{2\pi}{n\ddot{g}(x^*)}} \left\{ 1 - \frac{3g^{(4)}(x^*)}{4!n\ddot{g}(x^*)^2} + \frac{5 \cdot 3 \cdot g^{(3)}(x^*)^2}{2(3!)^2 n \ddot{g}(x^*)^3} + \mathcal{O}(n^{-2}) \right\}.$$

Simplifying coefficients gives the result. ■

Example 2.1 (Stirling's Approximation). Consider approximating $n!$ for $n \rightarrow \infty$. Express the factorial in terms of the Gamma function:

$$n! = \Gamma(n+1) = \int_0^\infty x^n e^{-x} dx = \int_0^\infty e^{n \ln(x) - x} dx.$$

Let $I = \int_0^\infty \exp\{n \ln(x) - x\} dx$. Make the change of variables $x = n\omega$, $dx = n d\omega$:

$$I = \int_0^\infty e^{n \ln(n\omega) - n\omega} n d\omega = n \int_0^\infty e^{n \ln(n) + n \ln(\omega) - n\omega} d\omega = n^{n+1} \int_0^\infty e^{-ng(\omega)} d\omega,$$

where $g(\omega) = \omega - \ln(\omega)$. The minimum of $g(\omega)$ occurs at $\dot{g}(\omega) = 1 - \omega^{-1} = 0$, or $\omega^* = 1$. Evaluating g and its derivatives at the critical point:

$$g(\omega^*) = 1, \quad \ddot{g}(\omega^*) = 1, \quad g^{(3)}(\omega^*) = -2, \quad g^{(4)}(\omega^*) = 6.$$

Applying (1.2.3) gives:

$$I = n^{n+1} e^n \sqrt{\frac{2\pi}{n}} \left\{ 1 - \frac{1}{8n} \cdot \frac{6}{1} + \frac{5}{24n} \left(\frac{-2}{1} \right)^2 + \mathcal{O}(n^{-2}) \right\}.$$

Simplifying coefficients gives:

$$n! = \Gamma(n+1) = \sqrt{2\pi} n^{n+1/2} e^n \left\{ 1 + \frac{1}{12n} + \mathcal{O}(n^{-2}) \right\},$$

for $n \rightarrow \infty$. ♠

Example 2.2 (Multivariate Laplace Approximation). Consider approximation of the integral:

$$I = \int_{\mathbb{R}^p} e^{-ng(\mathbf{x})} d\mathbf{x},$$

where $g(\mathbf{x})$ achieves a global minimum at $\mathbf{x}^*$. Take the Taylor expansion of g about $\mathbf{x}^*$:

$$g(\mathbf{x}) = g(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)' \ddot{g}(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + R(\mathbf{x}, \mathbf{x}^*).$$

Here $\ddot{g}(\mathbf{x})$ denotes the $p \times p$ Hessian:

$$\ddot{g}(\mathbf{x}) = \frac{\partial^2 g(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}'},$$

and the remainder $R(\mathbf{x}, \mathbf{x}^*) = \mathcal{O}(\|\mathbf{x} - \mathbf{x}^*\|^3)$.

Substituting the Taylor expansion into I :

$$\begin{aligned} I &= e^{-ng(\mathbf{x}^*)} \int e^{-(\mathbf{x} - \mathbf{x}^*)' n \ddot{g}(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*)/2} e^{-nR} d\mathbf{x} \\ &= e^{-ng(\mathbf{x}^*)} \int e^{-(\mathbf{x} - \mathbf{x}^*)' n \ddot{g}(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*)/2} \{1 + \mathcal{O}(nR)\} d\mathbf{x} \end{aligned}$$

Recall from multivariate normal distribution that:

$$\int e^{-(\mathbf{y}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu})/2} d\mathbf{y} = (2\pi)^{p/2} \det(\boldsymbol{\Sigma})^{1/2}.$$

Applied to integrate the leading term of I :

$$\int_{\mathbb{R}^p} e^{-ng(\mathbf{x})} d\mathbf{x} = (2\pi)^{p/2} e^{-ng(\mathbf{x}^*)} n^{-p/2} \det \{ \ddot{g}(\mathbf{x}^*) \}^{-1/2} \{ 1 + \mathcal{O}(n\|\mathbf{x} - \mathbf{x}^*\|^3) \}. \quad (2.2)$$

for $n \rightarrow \infty$. ♠

Example 2.3 (BIC). Consider the marginal likelihood of the data $f(\mathbf{y})$ in a Bayesian model:

$$f(\mathbf{y}) = \int f(\mathbf{y}|\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta} = \int e^{-ng(\boldsymbol{\theta})} d\boldsymbol{\theta},$$

where:

$$g(\boldsymbol{\theta}) = -\frac{1}{n} \ln f(\mathbf{y}|\boldsymbol{\theta}) - \frac{1}{n} \ln f(\boldsymbol{\theta}).$$

Applying (1.2.4):

$$f(\mathbf{y}) \propto f(\mathbf{y}|\boldsymbol{\theta}^*) f(\boldsymbol{\theta}^*) n^{-p/2} \det \{ \mathcal{J}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}^*) + \mathcal{I}_{\boldsymbol{\theta}\boldsymbol{\theta}'}^0(\boldsymbol{\theta}^*) \}^{-1/2}.$$

where:

$$\mathcal{J}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}) = -\frac{1}{n} \frac{\partial^2 \ln f(\mathbf{y}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}, \quad \mathcal{I}_{\boldsymbol{\theta}\boldsymbol{\theta}'}^0(\boldsymbol{\theta}) = -\frac{1}{n} \frac{\partial^2 \ln f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'},$$

are the observed and prior information for $\boldsymbol{\theta}$, respectively. Taking the logarithm:

$$\ln f(\mathbf{y}) \propto \ln f(\mathbf{y}|\boldsymbol{\theta}^*) + \ln f(\boldsymbol{\theta}^*) - \frac{p}{2} \ln(n) - \frac{1}{2} \ln \det \{ \mathcal{J}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}^*) + \mathcal{I}_{\boldsymbol{\theta}\boldsymbol{\theta}'}^0(\boldsymbol{\theta}^*) \}.$$

For increasing n , $\ln f(\boldsymbol{\theta}^*)$ and $\mathcal{I}_{\boldsymbol{\theta}\boldsymbol{\theta}'}^0$ are likely negligible, and $\mathcal{J}_{\boldsymbol{\theta}\boldsymbol{\theta}'}$ is expected to converge in probability to a constant. Hence the log marginal likelihood is roughly:

$$\ln f(\mathbf{y}) \approx \ln f(\mathbf{y}|\boldsymbol{\theta}^*) - \frac{p}{2} \ln(n).$$

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